ECE 171: Lab 5 Differential Amplifier

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1 Introduction

In this lab we were tasked with creating a Widlar current mirror and observing its various values in differential and common-mode.

2 Design

To start we constructed the differential amplifier with a current mirror with the CA3086 chip. Due to the imperfections of real life, such as the breadboard's open design, inaccurate resistance, the resistance of the measuring instruments, and other parasitic effects, we observed non-ideal results. We decided to then use a simulation to take data and gain a clearer conceptual understanding of the intended results.

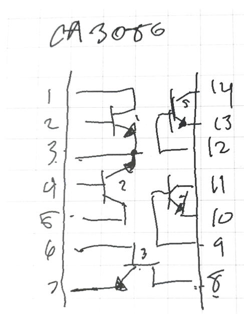


Figure 1: CA3086 Pin Out

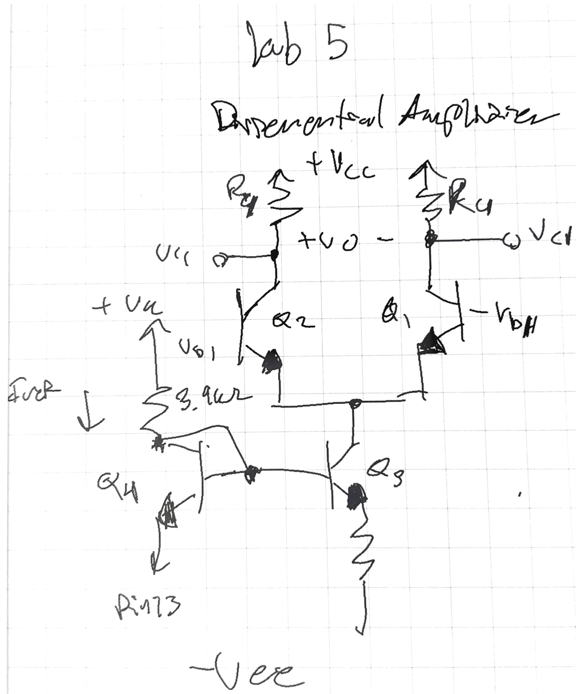


Figure 2: Circuit Schematic

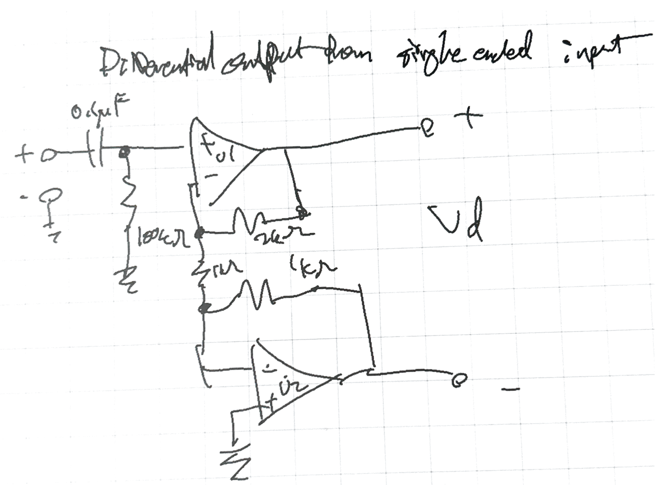


Figure 3: Circuit Addition Schematic

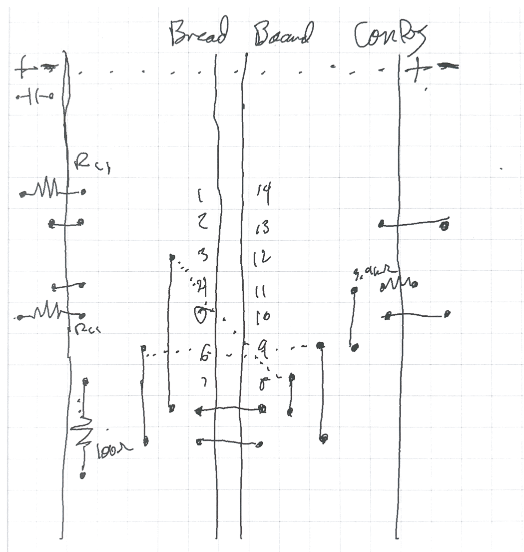


Figure 4: Preliminary Breadboard Configuration

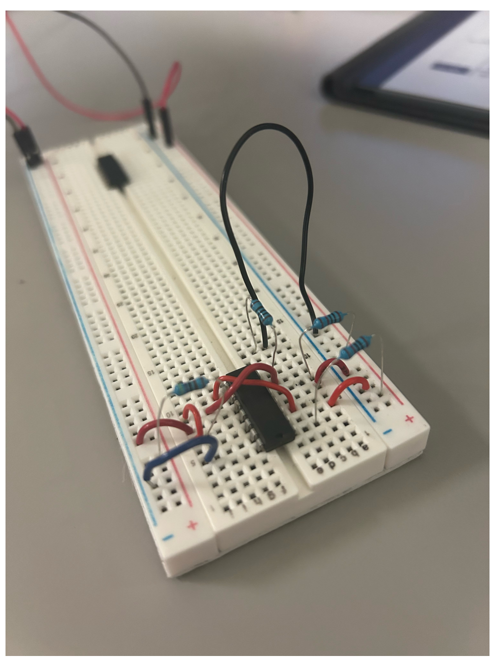


Figure 5: Preliminary Circuit

3 Part 1

In the first part of the lab we measured the voltage at the output terminals (the collector nodes) Vc3 and Vc4, and the reference voltage Vref. USing Ohms law we then calculated Iref, Ic3 and Ic4. Additionally we found the output voltage Vo from the difference between Vc3 and Vc4, which can be used to find Io.

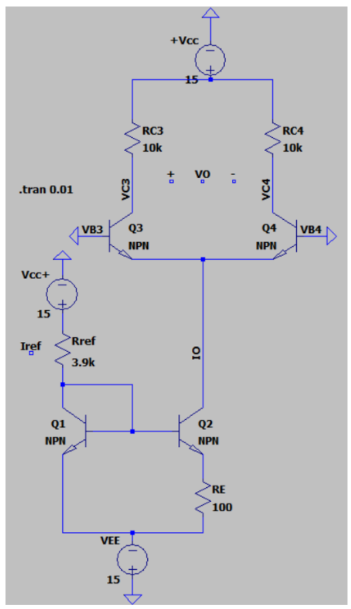


Figure 6: Part 1 Simulation Circuit

Table 1: Part 1 Collected Data

PART 1	**RC3**	**RC4**	**VC3**	**VC4**	**IC3**	**IC4**	**Rref**	**VREF**	**IREF**	**RE**	**V_RE = VC2**	**IE**	**IC = IE(B/B+1)**	**IO = IC**
Vcc = 15V Vee = 0V	10000	10000	3.1219	3.1219	0.00031219	0.00031219	3900	29.174	0.007480512821	100	0.063694	0.00063694	0.0006306336634	0.0006306336634

4 Part 2

In the second part we determined the quiescent collector voltages Vc3 and Vc4, and DC output offset voltage. Since we are using a simulation it is an ideal circuit and it is both expected and observed that the output voltage would be zero, which is the intended behavior since the voltages Vc3 and Vc4 ,whose difference gives the output voltage, should be equal to each other. The line of best fit would be a horizontal line at 0.

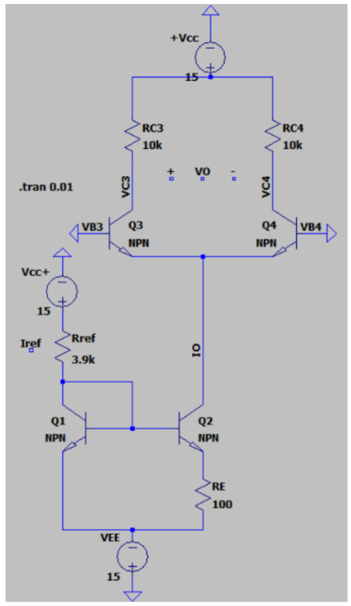


Figure 7: Part 2 Simulation Circuit

Table 2: Part 2 Collected Data

PART 2	VC3	VC4	DC Offset [V], VO
ideal	3.1219	3.1219	0

5 Part 3

In part 3 we measured the common mode gain. To set up this experiment we replaced the base voltages, which we grounded, with an AC input voltage Vcm to drive the amplifier. We then took the Vc3, Vc4 measurements. Due to the ideal circuit, the intended results of a gain of 1 to make the circuit a voltage follower, in the common mode.

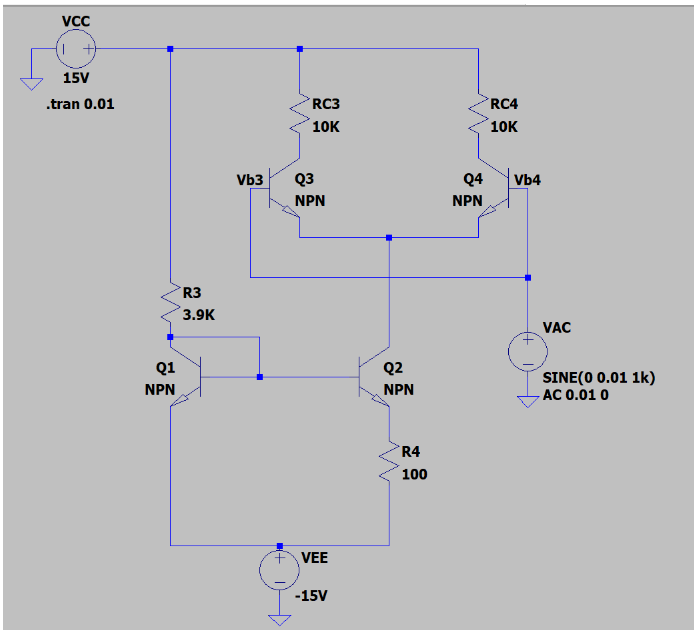
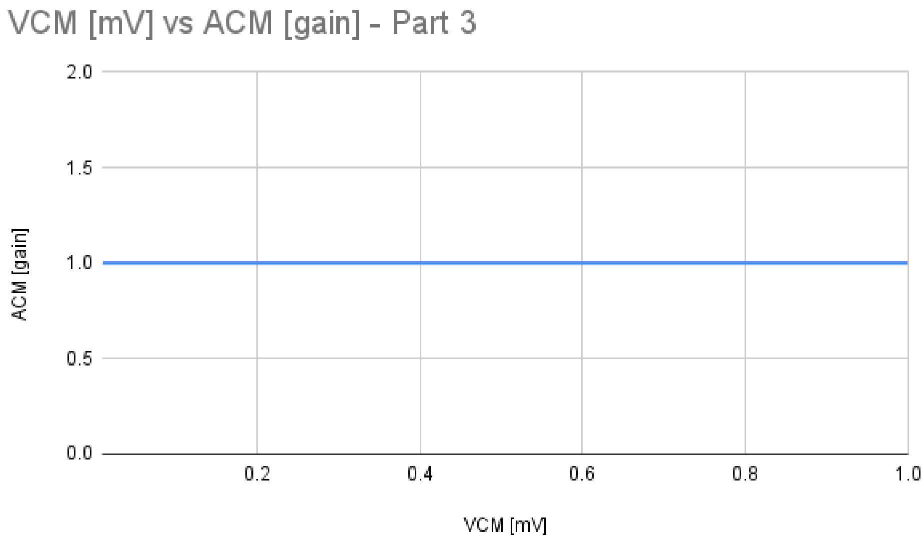


Figure 8: Part 3 Simulation Circuit

Table 3: Part 3 Collected Data

PART 3	Frequency [Hz]	VCM [mV]	VC3	VC4	VO	ACM (gain, dB)
w/ V (not pot)	1000	0.01	11.878	11.878	0	1
ampl=1-5mV	1000	0.03	11.878	11.878	0	1
	1000	0.07	11.878	11.878	0	1
	1000	0.1	11.878	11.878	0	1
	1000	0.25	11.878	11.878	0	1
	1000	0.4	11.878	11.878	0	1
	1000	0.55	11.878	11.878	0	1
	1000	0.7	11.878	11.878	0	1
	1000	0.85	11.878	11.878	0	1
	1000	1	11.878	11.878	0	1



6 Part 4

For part 4 we did the same as part 3 but used a DC input for Vcm, and observed the same values. Theoretically, all the voltages should vary, as using a potentiometer with a variable resistance changes every value in the circuit, with the excpetion of the collector voltages Vc3 and Vc4. Additionally, Vb3 and Vb4 should equal each other, same with Vc3 and Vc4.

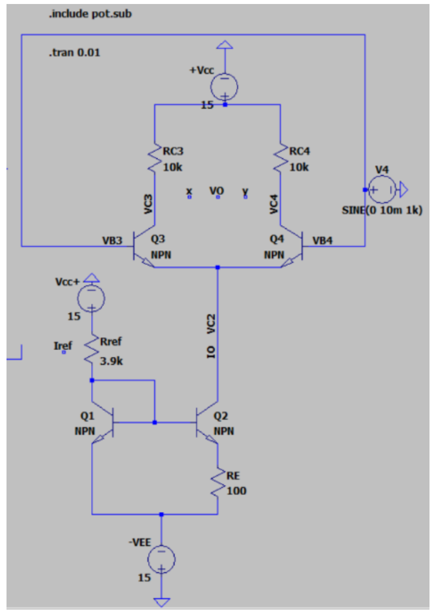


Figure 9: Part 3 Simulation Circuit

Table 4: Part 4 Collected Data

PART 4	**VCM [V] - base of pot**	**VC2 - where IO is**	**VBE3 - just B**	**VBE4**	**VC3 - just C**	**VC4**
wiper=0.035	0.52289	-0.22123	0.52289	0.52289	11.878	11.878
wiper=0.05	0.74703	0.0029131	0.74703	0.74703	11.878	11.878
wiper=0.065	0.97121	0.22708	0.97121	0.97121	11.878	11.878
wiper=0.085	1.2701	0.52602	1.2701	1.2701	11.878	11.878
wiper=0.1	1.4944	0.75026	1.4944	1.4944	11.878	11.878
when wiper=1 (100%)	14.98	14.222	14.98	14.98	14.244	14.244

7 Part 5

We added the special op-amp circuit to find the differential mode gain. The additional circuit element provided a true small-signal input. We can then determine the linear and non-linear regions of amplification. From experimentation, the graph of Vd and I was not informative, but a graph that compared Vd and the overall gain Ad was. Looking at the trend the current did not deviate much, but the gain did. A graph of Vd vs Ad shows that while the gain fluctuates a tad, there is an approximate exponential decay trend that shows where a linear and non-linear region with a cutoff point of around Vd = 0.0015.

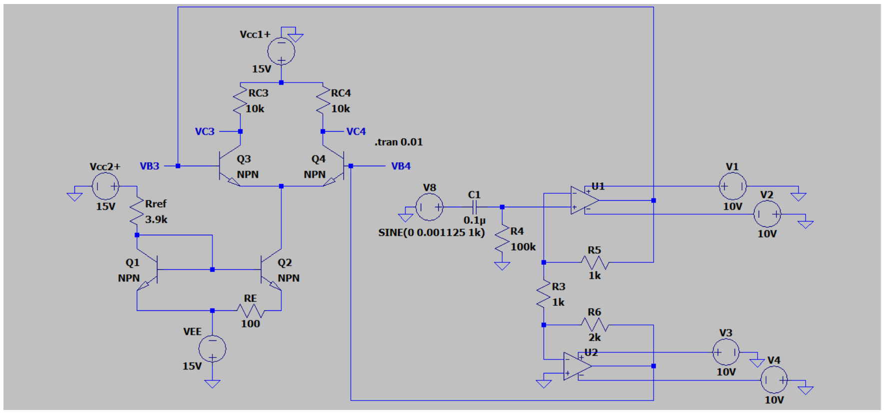


Figure 10: Part 5 Simulation Circuit

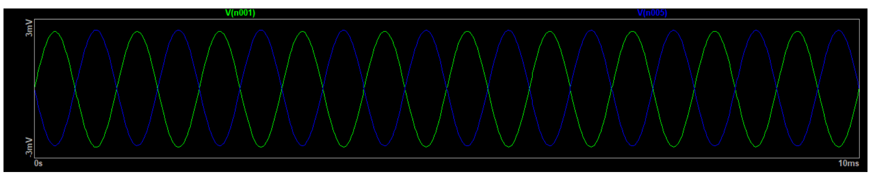


Figure 11: Part 5 LTSpice Output

Table 5: Part 5 Collected Data

Vin [V]	**vd [V]**	**VC3 [V]**	**VC4 [V]**	**ic3 [A]**	**ic4 [A]**	**Vo [V]**	**Ad**
0.00125	0.005	11.883	11.877	0.0011883	0.0011877	0.006	1.2
0.001125	0.0045	11.882	11.877	0.0011882	0.0011877	0.005	1.111111111
0.001	0.004	11.882	11.877	0.0011882	0.0011877	0.005	1.25
0.000875	0.0032	11.881	11.877	0.0011881	0.0011877	0.004	1.25
0.00075	0.003	11.88	11.877	0.001188	0.0011877	0.003	1
0.000625	0.0024	11.88	11.877	0.001188	0.0011877	0.003	1.25
0.0005	0.002	11.88	11.877	0.001188	0.0011877	0.003	1.5
0.000375	0.0015	11.879	11.877	0.0011879	0.0011877	0.002	1.333333333
0.00025	0.001	11.875	11.878	0.0011875	0.0011878	0.003	3

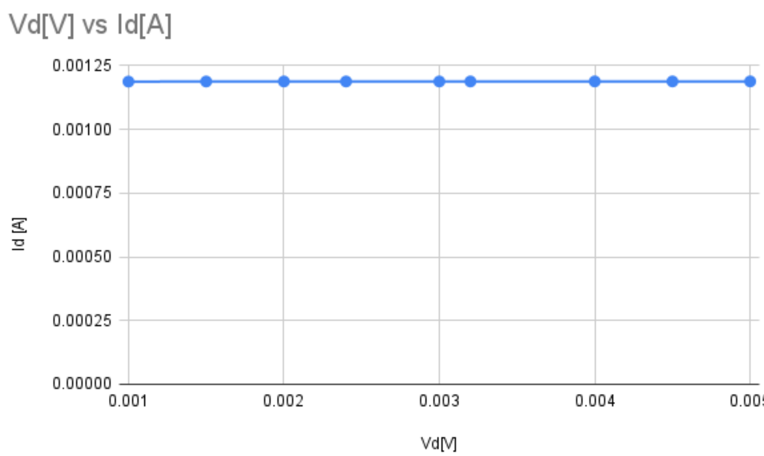


Figure 12: Flat Current

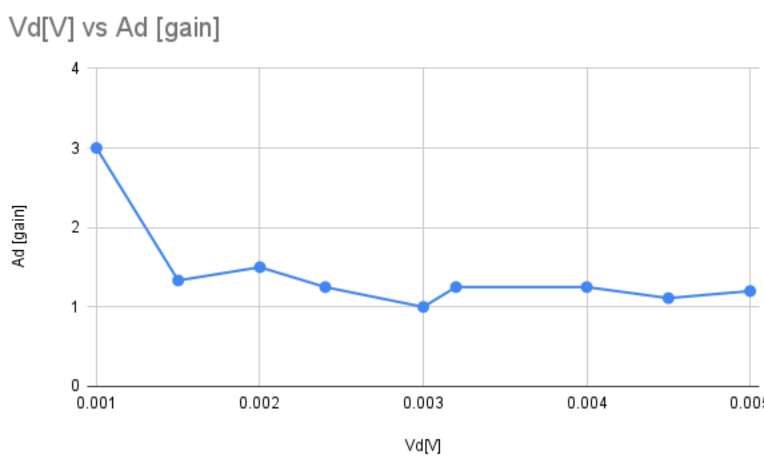


Figure 13: Exponential Decay Gain